

GRMHD Tutorial Day 3

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Part 3: What is GRMHD?

Outline

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

Outline

1. Euler and Einstein

2.

3.

4.

5.

6.

Outline

1. Euler and Einstein
2. Magnetic Field Additions
- 3.
- 4.
- 5.
- 6.

Outline

1. Euler and Einstein
2. Magnetic Field Additions
3. GRMHD!
- 4.
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4. Evolving the magnetic field
- 5.
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5. Coordinate systems
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1. Euler and Einstein
2. Magnetic Field Additions
3. GRMHD!
4. Evolving the magnetic field
5. Coordinate systems
6. Note on sound speed

Euler Equations

$$\begin{aligned}\frac{\partial \rho}{\partial t} &= -\nabla \cdot (\rho \vec{u}) \\ \frac{\partial E}{\partial t} &= -\nabla \cdot [(E + p)\vec{u}] \\ \frac{\partial(\rho \vec{u})}{\partial t} &= -\rho \vec{u} \cdot \nabla \vec{u} - \nabla p\end{aligned}$$

Euler Equations

$$\mathbf{U} = (\rho, E, \rho u^i)$$

$$\mathbf{F}^j = \begin{pmatrix} \rho u^j \\ u^j \rho u^i + p \delta^{ij} \\ (E + p) u^j \end{pmatrix}$$

$$\mathbf{P} = (\rho, u, \vec{u})$$

GRHD Equations

$$\begin{aligned}\partial_t (\sqrt{-g} \rho u^t) &= -\partial_j (\sqrt{-g} \rho u^j) \\ \partial_t (\sqrt{-g} T_\nu^t) &= -\partial_j (\sqrt{-g} T_\nu^j) + \sqrt{-g} T_\lambda^\kappa \Gamma_{\nu\kappa}^\lambda\end{aligned}$$

$$\mathbf{T}_{\text{HD}}^{\mu\nu} = (\rho + u + p) u^\mu u^\nu + p g^{\mu\nu}$$

U, F change!

$$\mathbf{U} = \sqrt{-g} (\rho u^t, T_t^t, T_i^t)$$

$$\mathbf{F}^j = \sqrt{-g} \begin{pmatrix} \rho u^j \\ T_t^j \\ T_i^j \end{pmatrix}$$

$$\mathbf{P} = (\rho, u, \vec{u})$$

Where $\sqrt{-g}$ is the root negative metric determinant, which we pre-compute at every zone center and face, as the metric will not change.

Adding Magnetic Fields

- $\vec{B}$ and $\vec{E}$ are just like in MHD! Electromagnetism is already relativistic
- I will not be deriving everything, but you can work this out from MTW if you define $B^i = F^{*it}$ from the Maxwell tensor and take $\vec{E}$ to vanish
- *But $\vec{E}$ only vanishes in fluid frame!*
 - We're purely working with the equations here, which are true in all frames!

Adding Magnetic Fields

The magnetic field also contains energy and momentum we need to track with the stress-energy tensor

$$T_{\text{EM}}^{\mu\nu} = b^2 u^\mu u^\nu + \frac{1}{2} b^2 g^{\mu\nu} - b^\mu b^\nu$$

GRMHD Equations:

$$\partial_t (\sqrt{-g} \rho u^t) = -\partial_i (\sqrt{-g} \rho u^i)$$

$$\partial_t (\sqrt{-g} T_\nu^t) = -\partial_i (\sqrt{-g} T_\nu^i) + \sqrt{-g} T_\lambda^\kappa \Gamma_{\nu\kappa}^\lambda$$

$$\partial_t (\sqrt{-g} B^i) = -\partial_j [\sqrt{-g} (b^j u^i - b^i u^j)]$$

$$\frac{1}{\sqrt{-g}} \partial_i (\sqrt{-g} B^i) = 0$$

$$T_{\text{MHD}}^{\mu\nu} = (\rho + u + p + b^2) u^\mu u^\nu + (p + \frac{1}{2} b^2) g^{\mu\nu} - b^\mu b^\nu$$

Final $\mathbf{U}, \mathbf{F}$

$$\mathbf{U} = \sqrt{-g} (\rho u^t, T_t^t, T_i^t, B^i)$$

$$\mathbf{F}^j = \sqrt{-g} \begin{pmatrix} \rho u^j \\ T_t^j \\ T_i^j \\ (b^j u^i - b^i u^j) \end{pmatrix}$$

$$\mathbf{P} = (\rho, u, \vec{u}, \vec{B})$$

Recovering primitive variables

- Determining $\mathbf{P}$ in terms of $\mathbf{U}$ is now not an analytic operation. But we update $\mathbf{U}$!
- Various solvers proposed:
 - Direct 5-way equation solve: slow and unreliable (Gammie et al. 2003)
 - Can be reduced to 1D: solve for enthalpy, determine Lorentz factor (Noble et al. 2006 $1D_W$)
 - By designing a 1D function, can guarantee a solution (Kastaun et al. 2020 "robust" or "RePrimAnd")
- There will always be $\mathbf{U}$ which do not correspond to any physical $\mathbf{P}$! That is, we cannot always find $\mathbf{P}$ so $\mathbf{U}(\mathbf{P}(\mathbf{U})) = \mathbf{U}$

Constraint equation

Recall our last GRMHD equation, $\frac{1}{\sqrt{-g}} \partial_i (\sqrt{-g} B^i) = 0$.

This is a constraint! It must be satisfied in initial conditions and throughout evolution.

Evolving the other equations *should* preserve this, but we're dealing with numerics here.

Constraint equation

Approaches:

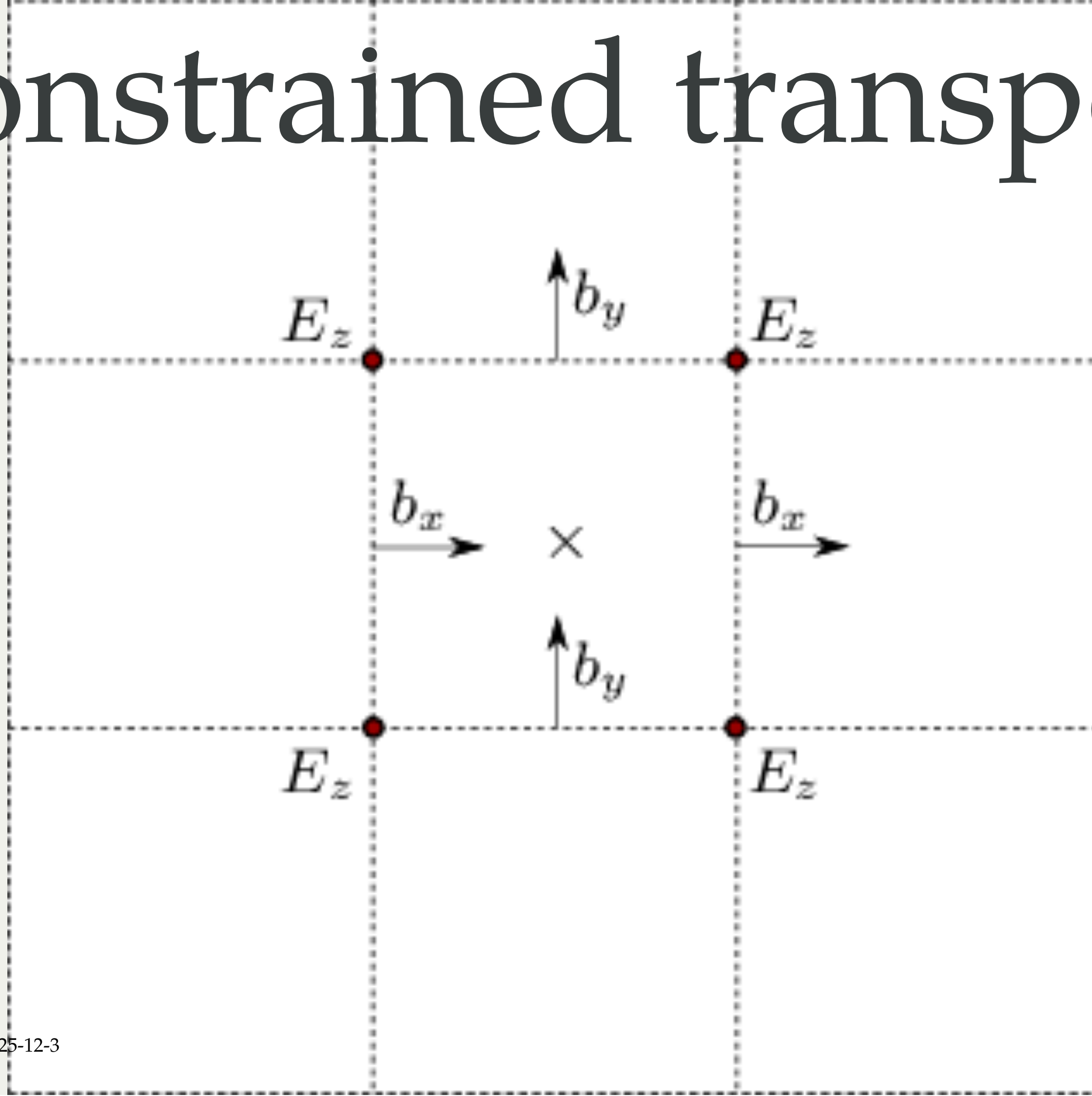
1. Just don't (bad idea)
2. "Clean up" any divergence every so often ("Elliptic maid")
3. Record constraint violations and advect/damp them away ("Constraint damping")
4. Revise the magnetic field evolution at each step to ensure the requirement is perfectly preserved ("Constrained transport")

Most static-spacetime GRMHD codes use #4!

Constraint equation

- Effectively, update the magnetic field using the current loops formed by the edges around each face
- Naturally this leads to defining B on the faces of each cell, and preserving the divergence at cell centers (Face-centered; Balsara & Spicer, Gardiner & Stone, etc.)
- However, simple constrained transport can be shifted to cell centers in each direction, for simplicity ("Flux-CT," Tóth 2000)

Constrained transport



Algorithm outline:

1. Consistent primitive and conserved variables $\mathbf{P}, \mathbf{U}$
2. Reconstruct values of primitive variables at the faces of zones
3. Calculate $\mathbf{U}_L, \mathbf{U}_R, \mathbf{F}_L, \mathbf{F}_R$
4. Riemann solve
5. Update $\mathbf{B}$ using constrained transport
6. Subtract $\nabla \cdot \mathbf{F}$ from $\mathbf{U}$ and add $T_{\lambda}^{\kappa} \Gamma_{\nu\kappa}^{\lambda}$
7. Compute new primitive variables
8. Floors, source terms, output, operator-split physics

Coordinate choices

- We can make arbitrary transformations to the metric $g^{\mu\nu}$!
- Simulation zones are evenly spaced in the transformed coordinate system
- This lets us move around simulation resolution by compressing the *coordinate system*!

Coordinate choices

e.g.,

$$r = \exp \left(X_{\text{FMKS}}^1 \right)$$

$$\theta = \theta_g + \exp \left(-s \left(X_{\text{FMKS}}^1 - X_{\text{FMKS},0}^1 \right) \right)$$

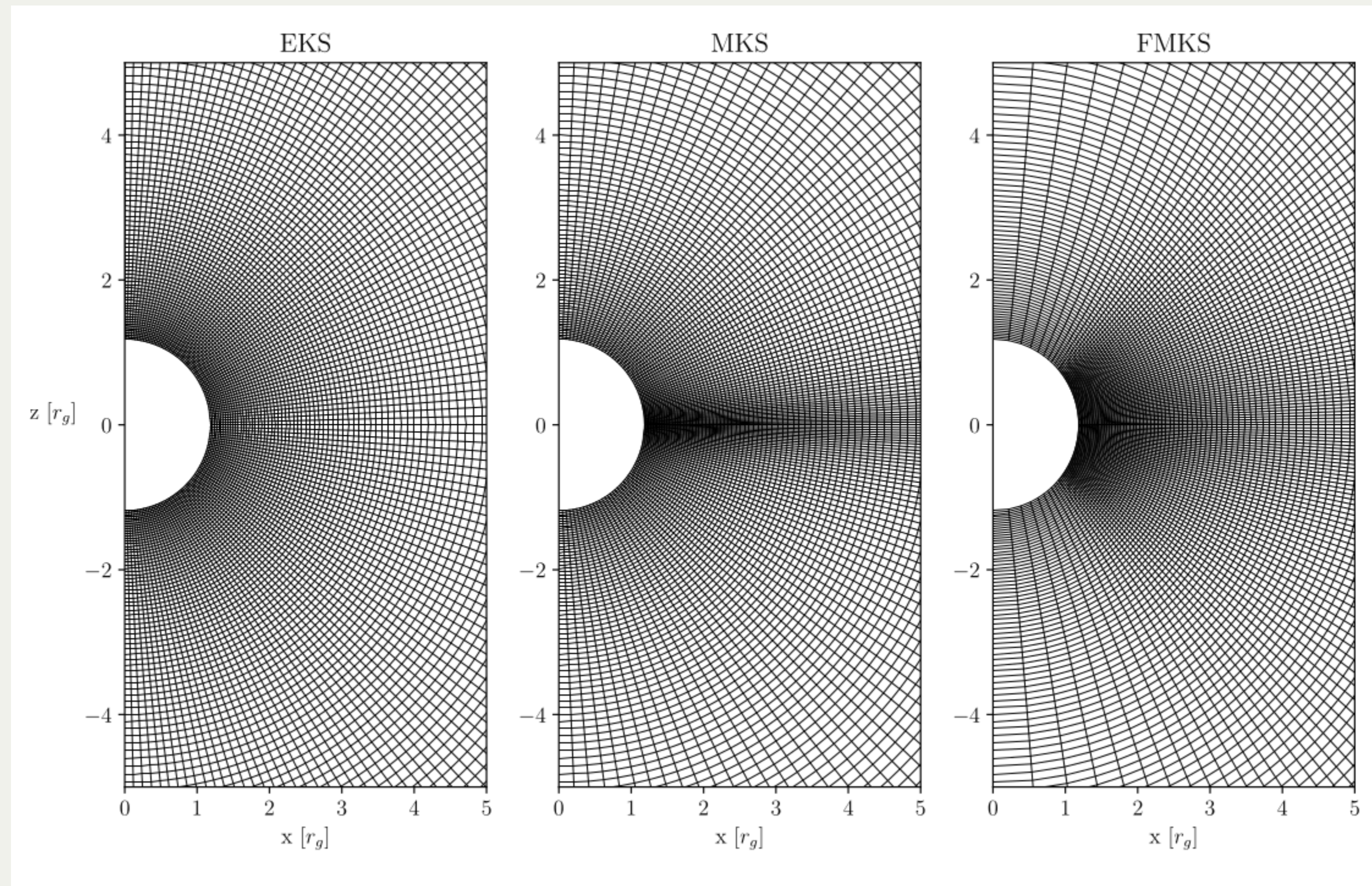
where

$$\theta_g = \pi X_{\text{FMKS}}^2 + \frac{1-h}{2} \sin \left(2\pi X_{\text{FMKS}}^2 \right)$$

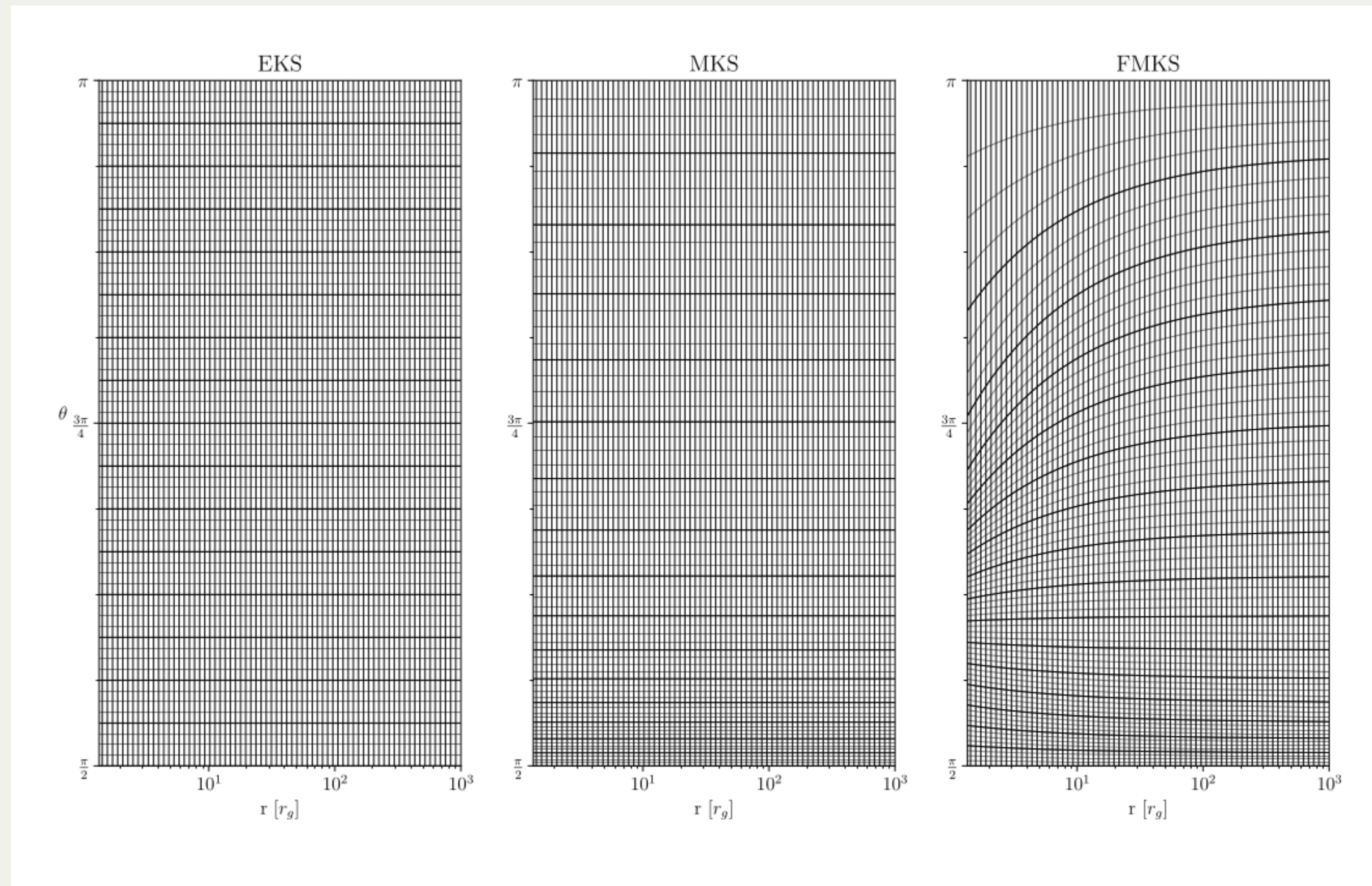
$$C = \frac{\pi}{2 + \frac{2}{A^B + B A^B}}$$

$$\theta = C \left(2 \sqrt{2} \left(1 + \left(\frac{2 X_{\text{FMKS}}^2 - 1}{A} \right)^B \right) \right) + \pi$$

Coordinate choices



Coordinate choices



Maximum wave speed ("sound" speed)

- Determined from the dispersion relation
- MHD version contains new oscillation modes (entropy, Alfvén, fast and slow magnetosonic)
- Full MHD dispersion relation is quartic, KHARMA uses a simpler quadratic form which overestimates
- GR adds *nothing*:
 - Composed of scalars
 - Speed is a scalar
- (However, if we wanted more waves we *would* need frame transformations)

Thanks!

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